

Trigonometry

Եռանկյունաչափություն

COURSE CODE	MTH-TR
PROGRAM	High School Mathematics · Ավագ դպրոցի մաթեմատիկա
LEVEL	Standalone course · typically Grades 10–12
DURATION	54–66 instructional hours (approx. 48 lessons)
PREREQUISITE	Algebra II (MTH-A2) or concurrent enrollment; Geometry recommended.
LEADS TO	Precalculus (MTH-PC) or Calculus (MTH-CA)

COURSE DESCRIPTION

Դասընթացի նկարագրություն

This standalone Trigonometry course develops the subject completely, from right triangle ratios through the unit circle, graphs, identities, equations, oblique triangles, polar coordinates, and vectors. It exists for three kinds of student: those whose school offers trigonometry as a separate semester course, those who need a focused treatment before Calculus because their Algebra II coverage was thin, and those preparing for competition or examination work where trigonometric fluency is decisive. It overlaps deliberately with Algebra II Unit 5 and Precalculus Units 3–4, but treats the material at greater depth.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

Trigonometry is the subject in which students most often accumulate a large collection of formulas with no organizing principle. Վարժարան teaches it from two anchors instead. The first is the unit circle, from which every value, every sign, every period, and the Pythagorean identity can be recovered. The second is the idea of the triangle as the tool for measuring the inaccessible, which is why the subject exists at all. A student who can rebuild the special values from a 30-60-90 triangle and who understands that all identities descend from a small number of relationships carries almost nothing in memory and still performs reliably.

PLACEMENT AND READINESS INDICATORS

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The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Solves right triangles using the Pythagorean theorem and simplifies radicals	The computational base of Units 1 and 5.
Graphs functions and applies transformations to a parent function	Prerequisite for Unit 3; without it, amplitude and phase shift become memorization.
Manipulates algebraic expressions including fractions with variables	Identity verification is algebraic manipulation applied to trigonometric expressions.
Understands the function concept and inverse functions	Prerequisite for inverse trigonometric functions in Unit 4.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

- 1 Define the six trigonometric functions as ratios in a right triangle and solve right triangles completely.
- 2 Convert between degree and radian measure and compute arc length and sector area.
- 3 Use the unit circle to extend the trigonometric functions to all real numbers and determine exact values.
- 4 Graph all six trigonometric functions and analyze amplitude, period, phase shift, and vertical shift.
- 5 Model periodic phenomena with sinusoidal functions and interpret every parameter in context.
- 6 Define and use the inverse trigonometric functions with their restricted domains.
- 7 Verify trigonometric identities and derive the sum, difference, double-angle, and half-angle formulas.
- 8 Solve trigonometric equations over a specified interval and in general form.
- 9 Apply the Law of Sines and the Law of Cosines, including the ambiguous case, and compute triangle areas.
- 10 Represent complex numbers in polar and trigonometric form and apply De Moivre's theorem.
- 11 Graph polar equations and convert between polar and rectangular coordinates.
- 12 Perform vector operations and resolve vectors into components in applied contexts.

COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 · 8-9 LESSONS

Right Triangle Trigonometry

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TOPICS COVERED

- The six trigonometric ratios and their reciprocal relationships
- Why the ratios depend only on the angle: the similarity argument
- Exact values for 30° , 45° , and 60° from the special triangles
- Solving right triangles for missing sides and angles
- Inverse trigonometric functions for finding an angle
- Cofunction relationships
- Angles of elevation and depression
- Bearings and navigation problems
- Multi-step applied problems requiring an auxiliary right triangle
- Accuracy, rounding, and significant figures in applied trigonometry

LEARNING OBJECTIVES

- 1 Solve any right triangle given sufficient information.
- 2 Derive the exact values for the special angles from the 30-60-90 and 45-45-90 triangles.
- 3 Set up and solve a multi-step applied problem requiring an auxiliary triangle.
- 4 Explain, using similarity, why the ratios are functions of the angle alone.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves right triangles at 92% accuracy.
- Produces exact special-angle values without reference material in 9 of 10 trials.
- Models and solves multi-step applied problems in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Sides are labelled opposite and adjacent without reference to a specific angle.	Mark the reference angle first. Relabel the same triangle from the other acute angle to make the dependence unmistakable.
The inverse function is confused with the reciprocal: $\sin^{-1} x$ taken as $1/\sin x$.	Contrast $\sin^{-1} x$ and $(\sin x)^{-1}$ explicitly in the same lesson and require correct notation thereafter.

MISCONCEPTION	RECOMMENDED RESPONSE
Intermediate results are rounded, and the accumulated error corrupts the final answer.	Carry full precision throughout and round only at the end. Demonstrate the divergence with a worked example.

TEACHING NOTE

Deriving the special values from the triangles rather than memorizing a table is the single most valuable habit in this unit. A student who can reconstruct them is never stranded.

UNIT 2 • 8-9 LESSONS**The Unit Circle and Radian Measure**

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TOPICS COVERED

- Angles in standard position; positive and negative rotation
- Coterminal angles and reference angles
- Radian measure as arc length over radius
- Converting between degrees and radians
- Arc length and sector area in radians
- Angular and linear velocity
- The unit circle definition of sine and cosine
- Extending all six functions to any angle
- Signs of the functions by quadrant
- Exact values at all special angles in all four quadrants
- Periodicity and its consequences
- Even and odd properties of the functions
- The Pythagorean identity derived from the unit circle
- Finding all six function values from one value and a quadrant

LEARNING OBJECTIVES

- 1 Convert fluently between degrees and radians and compute arc length and sector area.
- 2 Determine exact values of all six functions at any special angle in any quadrant.
- 3 Derive the Pythagorean identity from the unit circle.
- 4 Given one function value and a quadrant, determine the other five.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Produces exact values in all four quadrants in 9 of 10 trials without reference material.
- Converts and computes with radians at 92% accuracy.
- Determines all six values from one value and a quadrant in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The reference angle is used without applying the correct sign for the quadrant.	Reference angle gives the magnitude; the quadrant gives the sign. Require both to be stated separately before combining.
Radians are treated as merely another unit rather than as a ratio.	A radian is arc length divided by radius — a pure number. This is why calculus formulas require radians, and saying so now prevents confusion later.
Negative angles are handled by ignoring the sign.	Direction of rotation matters. Draw the angle before evaluating anything.

TEACHING NOTE

Require the student to build the unit circle from scratch several times across the unit, reasoning from the special triangles rather than recalling positions. The goal is reconstruction ability, not recall.

UNIT 3 • 8-9 LESSONS**Graphs of Trigonometric Functions**

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TOPICS COVERED

- Graphs of sine and cosine from the unit circle
- Amplitude, period, phase shift, and vertical shift
- The effect of each parameter
- Graphing $y = a \cdot \sin(b(x - c)) + d$ and the cosine analogue
- Writing an equation from a sinusoidal graph
- Graphs of tangent and cotangent; asymptotes and period
- Graphs of secant and cosecant related to their reciprocals
- Modelling periodic phenomena: tides, daylight, temperature, circular motion, sound
- Choosing sine or cosine for a model and justifying the choice
- Damped and combined trigonometric functions introduced informally
- Reading a periodic data set and fitting a sinusoid

LEARNING OBJECTIVES

- 1 Graph any of the six functions with all transformations applied correctly.
- 2 Write an equation from a graph or from a described periodic situation.
- 3 Interpret every parameter of a sinusoidal model in the context it describes.
- 4 Fit a sinusoidal model to real periodic data and assess the fit.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Graphs all transformed functions at 90% accuracy.
- Writes equations from graphs in 9 of 10 trials.
- Interprets all four model parameters in context in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Phase shift is read without factoring out the horizontal coefficient.	Factor first. In $\sin(3x - \pi)$, the shift is $\pi/3$. This is the most common graphing error in the course.
Period is confused with frequency.	Period is $2\pi/b$; frequency is its reciprocal. Compute both and state their units.
Asymptotes of the tangent are placed at multiples of π rather than at odd multiples of $\pi/2$.	The tangent is undefined where the cosine is zero. Derive the asymptote locations from that fact rather than memorizing them.

TEACHING NOTE

Modelling with real data — tide tables, sunrise times, temperature records — makes this unit concrete and is the strongest available argument for why periodic functions are studied at all.

UNIT 4 • 9-10 LESSONS**Identities and Trigonometric Equations**

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TOPICS COVERED

- Fundamental identities: reciprocal, quotient, Pythagorean, even-odd, cofunction
- Strategies for verifying an identity
- Simplifying trigonometric expressions
- Sum and difference formulas and their derivation
- Double-angle formulas and the three forms of the cosine double angle
- Half-angle formulas
- Product-to-sum and sum-to-product formulas
- Inverse trigonometric functions and their restricted domains and ranges
- Compositions involving inverse trigonometric functions
- Solving basic trigonometric equations over an interval
- Solving equations requiring an identity first
- Solving equations with multiple angles
- Writing the general solution of a trigonometric equation
- Extraneous solutions arising from squaring

LEARNING OBJECTIVES

- 1 Verify a trigonometric identity using a systematic strategy rather than trial and error.
- 2 Derive the double-angle formulas from the sum formulas.
- 3 Solve trigonometric equations over an interval and write the general solution.
- 4 Evaluate compositions involving inverse trigonometric functions with correct domain reasoning.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Verifies identities of moderate difficulty in 4 of 5 tasks.
- Solves trigonometric equations, finding all solutions, at 88% accuracy.

- Evaluates inverse compositions correctly in 8 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Both sides of an identity are operated on as if it were an equation.	An identity is verified by transforming one side into the other. Working on both sides assumes what is to be proved.
Solutions outside the range of the inverse function are reported.	The inverse functions have restricted ranges by definition. State the range before evaluating.
Solutions lost when dividing both sides by a trigonometric factor.	Factor rather than divide. Dividing by $\cos x$ discards every solution where $\cos x = 0$.

TEACHING NOTE

Identity verification is where students most often flail. Teach three explicit strategies — convert everything to sine and cosine, work with the more complicated side, look for a Pythagorean substitution — and require the student to name the strategy before starting.

UNIT 5 · 6-7 LESSONS

Oblique Triangles and Applications

Թեք եռանկյուններ և կիրառություններ

TOPICS COVERED

- The Law of Sines and its derivation
- The ambiguous case (SSA) analyzed completely
- The Law of Cosines and its derivation
- Choosing between the two laws
- Area of a triangle using two sides and the included angle
- Heron's formula
- Applied problems: surveying, navigation, force resolution
- Multi-triangle problems requiring a sequence of solutions
- Accuracy and the propagation of measurement error

LEARNING OBJECTIVES

- 1 Derive both laws and choose correctly between them for a given configuration.
- 2 Analyze the ambiguous case completely, determining zero, one, or two triangles.
- 3 Compute triangle area by two different methods and choose appropriately.
- 4 Solve a multi-triangle applied problem with a clearly stated diagram.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Chooses the correct law in 9 of 10 configurations.
- Resolves the ambiguous case correctly in 4 of 5 trials.
- Solves multi-triangle applied problems in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The ambiguous case is not recognized, and only one triangle is reported.	Whenever SSA is given, check for a second solution explicitly. Make it a written checklist step.
The Law of Sines is used where the Law of Cosines is required.	The Law of Sines needs an angle-side pair. If none is available (SSS or SAS), the Law of Cosines is required. Identify the given configuration before selecting.

TEACHING NOTE

The ambiguous case is worth a full lesson with physical construction. Students who have built two different triangles from the same SSA data never forget to check for the second one.

UNIT 6 • 8-9 LESSONS

Polar Coordinates, Complex Numbers, and Vectors

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TOPICS COVERED

- The polar coordinate system and multiple representations of a point
- Converting between polar and rectangular coordinates
- Graphing polar equations: circles, lines, cardioids, limaçons, roses, lemniscates
- Symmetry tests for polar graphs
- Trigonometric (polar) form of a complex number; modulus and argument
- Multiplication and division in trigonometric form
- De Moivre's theorem and powers of complex numbers
- Roots of complex numbers and their geometric arrangement
- Vectors: magnitude, direction, and component form
- Vector addition, subtraction, and scalar multiplication
- The dot product and the angle between vectors
- Applications: resultant force, velocity, work, and navigation

LEARNING OBJECTIVES

- 1 Convert between polar and rectangular forms for both points and equations.
- 2 Graph a polar equation and identify its symmetry.
- 3 Apply De Moivre's theorem to compute powers and roots of complex numbers.
- 4 Resolve a vector into components and solve an applied force or velocity problem.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Converts between coordinate systems at 92% accuracy.
- Applies De Moivre's theorem correctly in 8 of 10 trials including roots.
- Solves applied vector problems in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
A polar point is assumed to have a unique representation.	Adding 2π to the angle, or negating r and adding π , gives the same point. Ask for three representations of the same point.
When finding n complex roots, only one is reported.	There are exactly n , equally spaced around a circle. Plot them; the geometric picture makes the count obvious.
Vectors are added by adding magnitudes.	Direction matters. Draw the parallelogram, then work in components.

TEACHING NOTE

This unit is the direct bridge to Precalculus and to the parametric and polar sections of AP Calculus BC. The geometric picture of complex roots is also one of the most beautiful results a secondary student encounters — do not present it merely as a computation.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Placement Assessment	Before enrollment	Algebra II function and graphing skills, radicals, and right triangle geometry.
Unit Check	End of each unit · 50 min	Unit mastery criteria including non-calculator exact-value work.
Unit Circle Practical	Weekly from Unit 2 · 5 min	Reconstruction of exact values without reference material; recorded as a running chart.
Identity Verification Portfolio	Continuous from Unit 4	The student's verified identities with strategies named; evidence of systematic rather than trial-and-error work.
Mid-Course Assessment	After Unit 3	Cumulative Units 1–3.
End-of-Course Assessment	After Unit 6	All course outcomes, with a required non-calculator section.

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Semester	2 lessons/week, 60 min	Full course in approximately 24 weeks — the standard track.
Standard Year	1 lesson/week, 60 min	Full course in approximately 48 weeks.
Intensive	3 lessons/week	Full course in approximately 16 weeks; common as pre-Calculus preparation.
Summer Program	4 lessons/week, 6 weeks	Units 1–4 — the core, for students entering Precalculus or Calculus in the autumn.

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE

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Fifty minutes per lesson: ten to fifteen problems. Every assignment includes a non-calculator exact-value section, because calculator dependence is the most common failure mode in trigonometry and it is invisible until an examination forbids the device. Identity verification problems require the student to name the strategy before executing it. Applied problems require a labelled diagram; an answer without a diagram is marked incomplete.

PROGRESS MILESTONES REPORTED TO PARENTS

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These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- ◆ Solves right triangles and derives exact special-angle values from the triangles.
- ◆ Converts between degrees and radians and computes arc length and sector area.
- ◆ Reconstructs the unit circle from first principles.
- ◆ Determines exact values of all six functions in all four quadrants.
- ◆ Graphs all six trigonometric functions with all transformations.
- ◆ Models periodic phenomena and interprets every parameter in context.
- ◆ Verifies identities using a named systematic strategy.
- ◆ Derives the sum, difference, double-angle, and half-angle formulas.
- ◆ Solves trigonometric equations over an interval and in general form.
- ◆ Applies the Law of Sines and Cosines, including the full ambiguous case.
- ◆ Works with polar coordinates and applies De Moivre's theorem.
- ◆ Performs vector operations and solves applied force and velocity problems.
- ◆ Completed Trigonometry — ready for Precalculus or Calculus.

MATERIALS AND RESOURCES

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- Graphing calculator with trigonometric and complex number capability.
- Student-constructed unit circle, rebuilt rather than referenced.
- Dynamic graphing software for exploring transformations and polar curves.
- Real periodic data sets (tide tables, sunrise times, temperature records) for modelling.
- Վարժարան Trigonometry problem bank — approximately 1,300 items with a 250-item non-calculator sub-bank.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.